Ideological Expansion and Paradox

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Table of Contents

Introduction

The Algebra of Propositions

We assume propositions form a Boolean algebra $\mathcal{A}$:

- There is a greatest proposition $\top$ and a least proposition $\bot$.
- Each proposition p has a complement $\sim p$.
- Two propositions p and q have a join $p \cup q$ and a meet $p \cap q$.
- Join and meet distribute over each other.

$\mathcal{A}$ is a Boolean algebra $(A, \cup, \cap, \sim, \top, \bot)$.

Quantified Propositional Logic

The Language of QPL

Let $\mathcal{L}$ be the language of quantified propositional logic:

	expressions	axioms
variables	$p, q, r, \ldots$	
constants	$\bot, \top$	
connectives	$\neg, \wedge, \vee$	classical tautologies
quantifiers	$\forall, \exists$	elementary quantification theory
identity	$=$	reflexivity, Leibniz's Law

A Boolean-Valued Semantics for QPL

Given a Boolean algebra $\mathcal{A}$:

$$\left(A, \cap, \cup, \sim, \top, \bot\right).$$

$$x \leq y \quad := \quad x \cap y = x$$

An assignment α maps propositional variables into elements of A:

$$\begin{array}{lcll}
[\![p]\!]_\alpha & := & \alpha(p) & \\
[\![\top]\!]_\alpha & := & \top & \text{top} \\
[\![\bot]\!]_\alpha & := & \bot & \text{bottom} \\
[\![\varphi \vee \psi]\!]_\alpha & := & [\![\varphi]\!]_\alpha \cup [\![\psi]\!]_\alpha & \text{join} \\
[\![\varphi \wedge \psi]\!]_\alpha & := & [\![\varphi]\!]_\alpha \cap [\![\psi]\!]_\alpha & \text{meet} \\
[\![\neg\varphi]\!]_\alpha & := & \sim [\![\varphi]\!]_\alpha & \text{complement}
\end{array}$$

On the assumption $\mathcal{A}$ is a **complete** Boolean algebra, we let:

$$[\![\exists p\ \varphi(p)]\!]_\alpha \quad := \quad \bigcup_{a \in A} [\![\varphi]\!]_{\alpha[a/p]}$$

Consistency

Complete Boolean algebras support Boolean-valued models of QPL interpreting existential quantification in terms of join.

We now expand $\mathcal{L}$ with a propositional operator Q. In line with [Kaplan, 1995], we interpret $Q\varphi$ as: φ *is uniquely queried*.

Ideological Expansion $\cdots$

We let the operator Q be governed by two simple axioms:

$$\begin{array}{ll} \forall p\ (Qp \neq \bot) & \text{Regularity} \\ \forall p \forall q\ (Qp \wedge Qq \rightarrow p = q) & \text{Relative Atomicity} \end{array}$$

In words:

- Whatever p may be, it is not absurd to uniquely query p.
- No two distinct propositions are uniquely queried.

Expand a Boolean algebra $(A, \leq)$ with a function f and set:

$$[\![Q\varphi]\!]_\alpha \quad := \quad f([\![\varphi]\!]_\alpha)$$

For $(A, \leq, f)$ to model QPL + Regularity + Relative Atomicity, it must satisfy two conditions:

Reg. $\forall x \in A, f(x) \neq \bot$.

RAt. $\forall x \in A \; \forall y \in A \; (x \neq y \rightarrow f(x) \cap f(y) = \bot)$

Call $(A, \leq, f)$ a *well-behaved* iff it satisfies both constraints.

Incompleteness

No well-behaved expansion $(A, \leq, f)$ of a Boolean algebra $(A, \leq)$ is complete.

If $(A, \leq, f)$ is a well-behaved expansion, then the range of f on A forms an *anti-chain*, i.e., a family of pairwise disjoint non-zero elements, of the same cardinality as A.

No two distinct subsets of the anti-chain share a join in the algebra. Since the anti-chain has the cardinality of A, not all of its subsets have a join in $(A, \leq, f)$.

If A is infinite, then there are no more $\mathcal{L}_Q$-definable subsets of A than elements of A. Yet ...

Non-closure under definable join

No well-behaved expansion $(A, \leq, f)$ of a Boolean algebra $(A, \leq)$ is closed under $\mathcal{L}_Q$-definable join.

Consider the condition:

$$Qp \wedge \neg p$$

No member of A is a join of instances $[\![Qp \wedge \neg p]\!]_{\alpha[a/p]}$ of the form:

$$\bigcup_{a \in A} (f(a) \cap \sim a).$$

Assume for reductio that some r is $\bigcup_{a \in A} (f(a) \cap \sim a)$.

- $f(r) \cap \sim r \leq r$ and $f(r) \cap \sim r \leq \sim r$.
- $f(r) \cap \sim r = \bot$, and

$$r = \bot \cup \bigcup_{a \neq r} (f(a) \cap \sim a)$$

- From $f(r) \cap \sim r = \bot$, we infer $f(r) \leq r$ and $f(r) = f(r) \cap r$. By **RAt**, $f(r) \ \cap \ f(a) = \bot$ for all $a \neq r$ and:

$$f(r) = f(r) \cap \bigcup_{a \neq r} (f(a) \cap \sim a) = \bot.$$

This contradicts **Reg** above.

Boolean unsatisfiability

No complete Boolean algebra supports a Boolean-valued model of QPL + Regularity + Relative Atomicity

This is not a limitation of the algebraic semantics, but rather a symptom of a more serious issue:

$$\begin{aligned}\bigcup_{a\in A} f(a) \cap \sim a \quad &:= \quad \bigcup_{a\in A} [\![Qp \wedge \neg p]\!]_{[a/p]} \\ &:= \quad [\![\exists p(Qp \wedge \neg p)]\!]\end{aligned}$$

But $\exists p(Qp \wedge \neg p)$ plays a crucial role in Prior's Theorem $\cdots$

Prior's Theorem

$\vdash Q\exists p(Qp \wedge \neg p) \rightarrow \exists p(Qp \wedge p) \wedge \exists p(Qp \wedge \neg p)$

Let λ be $\exists p(Qp \wedge \neg p)$. Given classical quantificational logic, $\vdash Q\lambda \rightarrow \lambda$, whence $\vdash Q\lambda \rightarrow (Q\lambda \wedge \lambda)$ and $\vdash Q\lambda \rightarrow \exists p(Qp \wedge p)$. On the other hand, $\vdash \lambda \rightarrow \exists p(Qp \wedge \neg p)$.

The existence of such a proposition as $\exists p(Qp \wedge \neg p)$ is inconsistent with Regularity and Relative Atomicity.

Inconsistency

QPL + Regularity + Relative Atomicity is inconsistent.

By Regularity, $Q\lambda \neq \bot$. By Prior's theorem, we have both $\exists p(Qp \wedge p) \neq \bot$ and $\exists p(Qp \wedge \neg p) \neq \bot$. But if μ and ν such that $(Q\mu \wedge \mu)$ and $(Q\nu \wedge \neg\nu)$, then $Q\mu \wedge Q\nu \neq \bot$ and $\mu \neq \nu$, which conflicts with Relative Atomicity.

This is closely related to Kaplan's paradox when we interpret $Q\varphi$ as: φ is *possibly* uniquely Q.

Two Responses: Expansion and Refinement

Ramification

One response, in line with [Kaplan, 1995], is to ramify:

- Propositions of level 0 concern *extensional matters*, e.g., distributions of earth, air, fire, and water.
- Propositions of level 1 concern both extensional matters and patterns of application of Q to propositions of level 0.
- Propositions of level 2 include propositions of level 1 and patterns of application of Q to level 1.

Here is how [Bacon et al., 2016] implement the proposal:

How to Model Ramification

Given a set W_0 of proto-worlds, we define:

$$
\begin{aligned}
\mathcal{S}_0 &= W_0 & \qquad \mathcal{I}_0 &= \mathcal{P}(\mathcal{S}_0) \\
\mathcal{S}_1 &= W_0 \times \mathcal{I}_0 & \qquad \mathcal{I}_1 &\subseteq \mathcal{P}(\mathcal{S}_1) \\
\\
\mathcal{S}_n &= W_0 \times \mathcal{I}_0 \times \cdots \times \mathcal{I}_{n-1} & \qquad \mathcal{I}_n &\subseteq \mathcal{P}(\mathcal{S}_n)
\end{aligned}
$$

$\mathcal{S}_n$ consists of stage n possibilities $(a, E_1, \ldots, E_{n-1})$, which:

- specify a proto-world a
- record which stage m intensions, for $m < n$, are Q. $E_m \subseteq \mathcal{I}_m$ is the set of stage m intensions that are Q.

Not all sets of stage n possibilities carry new information not encoded by earlier sets, e.g., $\mathcal{S}_1$ carries no information beyond $\mathcal{S}_0$.

Stage n intensions are sets of stage n possibilities, which carry *new* information not encoded by earlier sets of possibilities.

A subset $X \subseteq \mathcal{S}_n$ is *generated at* stage $m \leq n$ iff for some $Y \subseteq \mathcal{S}_m$ $X = \{s \in \mathcal{S}_n : s{\restriction}m \in Y\}$.

$$\mathcal{I}_n \quad = \quad \{X \subseteq \mathcal{S}_n : X \text{ not generated at some earlier stage}\}$$

Worlds and Completions

We now let a *world* w be an ω-sequence of the form:

$$(a, E_0, \ldots, E_n, \ldots),$$

where a is a proto-world and E_n is a set of stage n intensions.

Call w a *completion* of a stage n possibility s iff $w \restriction n = s$.

W is the set of all complete finite stage possibilities.

Propositions and Completions

We let the *completion* $\widehat{I}$ of a stage n possibility X be the set of world completions of stage n possibilities in X.

$$\widehat{X} = \{w : w{\restriction}n \in X\}$$

Every completion of a set $X \subseteq \mathcal{S}_n$ is a completed stage m intension for some $m \leq n$.

If $\widehat{I}$ is the completion of a stage n intension I, then:

- $\widehat{I}$ agrees with I up to stage n.
- $\widehat{I}$ is noncommittal after stage n.

Completions are closed under Boolean operations.

Propositions

We have a tower of Boolean algebras

$$(A_n, \cap_n, \cup_n, \sim_n, \varnothing, W)$$

where A_n is the set of completions of stage $\leq n$ intensions, and for each $n \in \omega$, $A_n \subset A_{n+1}$.

We have ramified propositions into cumulative levels: propositions of level n are completions of stage m intensions for $m \leq n$.

An assignment α maps propositional variables p_n of level n to completions of stage $\leq n$ intensions.

$[\![p_n]\!]^n_\alpha$	$:=$	$\alpha(p_n)$	
$[\![\top]\!]^n_\alpha$	$:=$	W	top
$[\![\bot]\!]^n_\alpha$	$:=$	$\emptyset$	bottom
$[\![\varphi \vee \psi]\!]^n_\alpha$	$:=$	$[\![\varphi]\!]^n_\alpha \cup [\![\psi]\!]^n_\alpha$	join
$[\![\varphi \wedge \psi]\!]^n_\alpha$	$:=$	$[\![\varphi]\!]^n_\alpha \cap [\![\psi]\!]^n_\alpha$	meet
$[\![\neg\varphi]\!]^n_\alpha$	$:=$	$\sim_n [\![\varphi]\!]^n_\alpha$	complement

We expand the Boolean algebras with a function f:

$$f(\widehat{I}) = \{w : I \in E_n\}$$

f maps a completion of a stage n intension to a set of worlds in which E_n contains the stage n intension. We now set:

$$[\![Q\varphi]\!]^n_\alpha \quad := \quad f([\![\varphi]\!]^n_\alpha)$$

Two observations:

- Regularity is satisfied, since $f(\widehat{I}) \neq \emptyset$ for each completion $\widehat{I}$.
- Relative Atomicity requires a restriction of W to worlds at which *all but at most one* E_n are empty and E_n is a singleton.

How to Block Prior

If $\alpha(p_n)$ is a completion $\widehat{I}$ of some stage $m \leq n$ intension, then:

$$\begin{aligned} [\![Qp_n \wedge \neg p_n]\!]^n_\alpha &= [\![Qp_n]\!]^n_\alpha \cap [\![\neg p_n]\!]^n_\alpha \\ &= f(\widehat{I}\,) \cap_n \sim_n \widehat{I} \\ &= \{w : I \in E_m\} \cap \{w : w{\restriction}m \notin I\} \\ &= \{w : I \in E_m \wedge w{\restriction}m \notin I\} \end{aligned}$$

For example:

$$\begin{aligned} [\![Qp_n \wedge \neg p_n]\!]^n_{\alpha[\varnothing/p_n]} &= \{w : \varnothing \in E_0 \wedge w{\restriction}0 \notin \varnothing\} \\ &= \{w : \varnothing \in E_0\} \end{aligned}$$

Existential quantification is interpreted in terms of join:

$$[\![\exists p_n(Qp_n \wedge \neg p_n)]\!]^n \quad = \quad \bigcup_{\widehat{I} \in A_n} \{w : I \in E_m \wedge w \restriction m \notin I\}$$

We let λ_n abbreviate: $\exists p_n(Qp_n \wedge \neg p_n)$.

$[\![\lambda_n]\!]^n$ is **not** a completion of a stage $m \leq n$ intension.

Since $\exists p_n(Qp_n \wedge \neg p_n)$ is **not** a proposition of level n, it falls outside the range of the quantifier $\exists p_n$ and the proof of Prior's theorem is blocked.

Suppose $[\![\lambda_n]\!]^n$ is the completion $\widehat{J}$ of some stage $m \leq n$ intension.
$[\![\lambda_n]\!]^n \neq \emptyset$, e.g., the world completion of $(a, \{\{a\}\})$ is in $[\![\lambda_n]\!]^n$.
Choose a world w for which

$$w{\restriction}m \in J \quad \text{and} \quad J \in E_m.$$

Is w a member of $[\![\lambda_n]\!]^n$?

$$\begin{aligned} w \in [\![\lambda_n]\!]^n &\Leftrightarrow w \in \widehat{J} \\ &\Leftrightarrow w{\restriction}m \in J \\ &\Leftrightarrow w{\restriction}m \in J \vee J \notin E_m \\ &\Leftrightarrow w \notin [\![\lambda_n]\!]^n \end{aligned}$$

Some observations:

- The proof of Prior's Theorem is blocked when we exclude λ_n from the range of the existential quantifier $\exists p_n$.
- The algebras in the tower of Boolean algebras

$$\left(A_n, \cap_n, \cup_n, \sim_n, \varnothing, W, f\right)$$

  are **not** closed under definable join.
- Regularity is satisfied.
- Relative Atomicity requires a further restriction on W.

We now look at a different strategy $\cdots$

Refinement

We now keep the domain constant, and look at successive refinements of the order. That is, we consider a series of pre-ordered sets of the form:

$$\Big(\bigcup_{n\in\omega} A_n, \sqsubseteq_n \Big)$$

For each stage n, define a preorder $\sqsubseteq_n$ on $\bigcup_{n\in\omega} A_n$:

$$\widehat{I} \sqsubseteq_n \widehat{J} \quad \text{iff} \quad \widehat{I}{\restriction}n \subseteq \widehat{J}{\restriction}n$$

Stage n Indiscernibility

The preorder determines an equivalence relation $\equiv_n$:

$$\widehat{I} \equiv_n \widehat{J} \quad \text{iff} \quad \widehat{I}\restriction n = \widehat{J}\restriction n$$

If $\widehat{I} \equiv_n \widehat{J}$, then they are *indiscernible* at stage n.

As we ascent into the hierarchy, we refine our powers of discrimination. At stage n, we remain blind to distinctions that only become apparent at later stages.

The Quotient Algebra

We have a Boolean algebra $(\bigcup_{n\in\omega} A_n/\equiv_n, \leq_n)$ for each stage n.

$$\begin{array}{lcll}
[\widehat{I}]_n \cup_n [\widehat{J}]_n & = & [\widehat{I} \cup \widehat{J}]_n & \text{join} \\
[\widehat{I}]_n \cap_n [\widehat{J}]_n & = & [\widehat{I} \cap \widehat{J}]_n & \text{meet} \\
\sim_n [\widehat{I}]_n & = & [\widehat{\sim I}]_n & \text{complement} \\
\top_n & = & [W]_n & \\
\bot_n & = & [\emptyset]_n &
\end{array}$$

We may choose a canonical representative for each equivalence class

$$c([\widehat{I}]_n) = \widehat{I{\restriction}n}.$$

$\widehat{I{\restriction}n}$ is a completion of some stage $m \leq n$ intension.

Individuation

A completion $\widehat{I}$ is **individuated at stage** n if it is identical to its canonical representative at that stage:

$$\widehat{I} \quad = \quad c([\widehat{I}]_n).$$

This is the case when $\widehat{I}$ is a completion of a stage n intension.

A completion $\widehat{I}$ is **individuated by stage** n iff $\widehat{I}$ is individuated at some earlier stage $m \leq n$.

Three observations:

- Both W and $\emptyset$ have been individuated at stage 0.
- Once $\widehat{I}$ is individuated at stage n, it remains individuated by a later stage.
- We argue that $\exists p(Qp \wedge \neg p)$ is never individuated.

Refinement Semantics

Given the pre-order:

$$(\bigcup_{n\in\omega} A_n, \sqsubseteq_n),$$

we let α map propositional variables into $\bigcup_{n\in\omega} A_n$:

$$\begin{array}{lcll}
[\![p]\!]^n_\alpha & = & \alpha(p) & \\
[\![\neg\varphi]\!]^n_\alpha & = & \sim_n c([\varphi]_n) & \text{complement} \\
[\![\varphi \vee \psi]\!]^n_\alpha & := & c([\varphi]_n) \cup_n c([\psi]_n) & \text{join} \\
[\![\varphi \wedge \psi]\!]^n_\alpha & := & c([\varphi]_n) \cap_n c([\psi]_n) & \text{meet} \\
[\![Q\varphi]\!]^n_\alpha & := & f(c([\varphi]_n)) &
\end{array}$$

Variables range over the constant domain, but the operations are sensitive only to the discernible behavior of a proposition.

Existential quantification is interpreted in terms of join.

$$[\![\exists p\ \varphi]\!]^n_\alpha \quad := \quad \bigcup_{\widehat{I}\ \in\ \bigcup_n A_n} c([\![\varphi]\!]^n_{\alpha[\widehat{I}/p]})$$

For the specific case:

$$[\![\exists p(Qp \wedge \neg p)]\!]^n \quad = \quad \bigcup_{\widehat{J|n}\ \in\ A_n} \{w : J{\restriction}m \in E_m \wedge w{\restriction}m \notin J{\restriction}m\}$$

where $\widehat{J{\restriction}n}$ is the completion of a stage m intension $J{\restriction}m$.

Let λ abbreviate $\exists p(Qp \wedge \neg p)$.

λ is never individuated

$[\![\lambda]\!]^n$ is **not** a completion of a stage $m \leq n$ intension.

Suppose $[\![\lambda]\!]^n$ is the completion $\widehat{J}$ of some stage $m \leq$ intension $\widehat{J}$. Choose a world w again for which

$$w{\restriction}m \in J \quad \text{and} \quad J \in E_m.$$

Is w a member of $[\![\lambda]\!]^n$?

$$\begin{aligned} w \in [\![\lambda]\!]^n \quad &\Leftrightarrow \quad w \in \widehat{J} \\ &\Leftrightarrow \quad w{\restriction}m \in J{\restriction}m \\ &\Leftrightarrow \quad w{\restriction}m \in J{\restriction}m \vee J{\restriction}m \notin E_m \\ &\Leftrightarrow \quad w \notin [\![\lambda]\!]^n \end{aligned}$$

If $\widehat{I}$ is individuated at stage n, then $f(\widehat{I{\restriction}n}) \equiv_n \top$

This is because:

$$f(\widehat{I{\restriction}n}) \quad = \quad \{w : I{\restriction}n \in E_n\}$$

Any stage n possibility $(a, E_0, \ldots, E_{n-1})$ may be extended into one in which $I{\restriction}n \in E_n$, e.g., $(a, E_0, \ldots, E_{n-1}, \{I{\restriction}n\})$. Therefore:

$$\begin{aligned} f(\widehat{I{\restriction}n}){\restriction}n &= \{w : w{\restriction}n \in \mathcal{S}_n\} \\ &= W{\restriction}n \end{aligned}$$

Thus:

$$f(\widehat{I{\restriction}n}) \quad \equiv_n \quad \top$$

Since neither $[\![\lambda]\!]^n$ nor $[\![\neg\lambda]\!]^n$ are individuated by stage n:

$$[\![Q\lambda]\!]^n \quad \equiv_n \quad \top \quad \equiv_n \quad [\![Q\neg\lambda]\!]^n$$

Two observations:

- $Q\lambda$ and $Q\neg\lambda$ are, relative to all finite stages, indiscernible form $\top$. This will help us vindicate Prior's Theorem.
- If $\widehat{I}$ is individuated at stage n, then we will have $[\![Qp \wedge Q\neg p]\!]^n_{\alpha[\widehat{I}/p]} \equiv_n \top$ even if we restrict the domain of worlds in order to accommodate RAt.

Prior Revisited

Recall Prior's Theorem:

$$Q\exists p(Qp \wedge \neg p) \rightarrow \exists p(Qp \wedge p) \wedge \exists p(Qp \wedge \neg p)$$

The antecedent $Q\lambda$ is n-indiscernible from $\top$ at all stages of refinement as is $Q\neg\lambda$. Since $\lambda \vee \neg\lambda$ is $\top$ relative to all stages of refinement, both $\exists p(Qp \wedge p)$ and $\exists p(Qp \wedge p)$ become indiscernible from $\top$ at every finite stage of refinement.

Both λ and $\neg\lambda$ provide witnesses to the existential generalizations in the consequent, but the reason is that neither $Q\lambda$ nor $Q\neg\lambda$ have come into focus at stage n.

- The pre-ordered sets

$$\Big(\bigcup_{n \in \omega} A_n, \sqsubseteq_n \Big)$$

are **closed under definable join**. When we compute the join of a definable set, we input the canonical representatives at stage n, which results in a completion of a stage $m \leq n+1$ intension.
The join in question may not yet individuated at stage n. That is exactly the situation with $[\![\lambda]\!]^n$, which is never individuated at stage n.

- Regularity is satisfied: $f(\widehat{I}) \neq \emptyset$ for all $\widehat{I}$ in $\bigcup_{n \in \omega} A_n$.
- Relative Atomicity collapses for propositions newly individuated at a given finite stage, but it is still satisfied by propositions individuated at strictly earlier stages when we restrict the domain of worlds to contain only worlds at which exactly one finite stage intension figures in some E_n.

Conclusion

We distinguish two approaches to the risks of ideological expansion.

A strategy of **ontological expansion** involves ramification and requires a restriction on existential generalization. Since λ_n lies outside the range of $\exists p_n$, we are not entitled to the application of existential generalization required for the proof of the theorem.

A strategy of **refinement** preserves classical logic and frames Prior's Theorem as a byproduct of unavoidable discriminatory limitations on our part. There is a fixed domain of propositions, but our ability to discriminate among them evolves through a succession of preorders, which correspond to ever more fine-grained relations of entailment on that domain.

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